Lab Assignment 1: Basic Array Operations

```
#include <stdio.h>

#include <stdlib.h>


int main() {

    int n, i, largest, smallest;

    float sum = 0, average;


    // Input the size of the array

    printf("Enter the number of elements in the array: ");

    scanf("%d", &n);


    // Declare an array of size n

    int *arr = (int *)malloc(n * sizeof(int));


    // Input elements from the user

    printf("Enter %d elements:\n", n);

    for (i = 0; i < n; i++) {

        scanf("%d", &arr[i]);

    }


    // Initialize largest and smallest elements

    largest = arr[0];

    smallest = arr[0];


    // Find the largest and smallest elements

    for (i = 1; i < n; i++) {

        if (arr[i] > largest) {
```

```c
        largest = arr[i];
    }
    if (arr[i] < smallest) {
        smallest = arr[i];
    }
}

// Sort the array in ascending order
for (i = 0; i < n - 1; i++) {
    for (int j = i + 1; j < n; j++) {
        if (arr[i] > arr[j]) {
            int temp = arr[i];
            arr[i] = arr[j];
            arr[j] = temp;
        }
    }
}

// Calculate the sum and average of the array elements
for (i = 0; i < n; i++) {
    sum += arr[i];
}
average = sum / n;

// Output the results
printf("Largest element: %d\n", largest);
printf("Smallest element: %d\n", smallest);
printf("Sorted array in ascending order: ");
for (i = 0; i < n; i++) {
```

```
        printf("%d ", arr[i]);
    }
    printf("\n");
    printf("Sum of elements: %.2f\n", sum);
    printf("Average of elements: %.2f\n", average);

    return 0;
}
```

```
Enter the number of elements in the array: 4
Enter 4 elements:
11
22
12
33
Largest element: 33
Smallest element: 11
Sorted array in ascending order: 11 12 22 33
Sum of elements: 78.00
Average of elements: 19.50
```

Lab Assignment 2: Array of Structures

```
#include <stdio.h>
#include <stdlib.h>

#define MAX_NAME_LEN 50

// Define the structure for Student
struct Student {
    char name[MAX_NAME_LEN];
    int age;
    float marks;
```

```
};

// Function prototypes
void inputStudentDetails(struct Student students[], int n);
void displayStudentDetails(struct Student students[], int n);
void sortStudentsByMarks(struct Student students[], int n);
struct Student findTopStudent(struct Student students[], int n);

int main() {
    int n;

    // Input number of students
    printf("Enter the number of students: ");
    scanf("%d", &n);

    // Declare an array of students
    struct Student *students = (struct Student *)malloc(n * sizeof(struct Student));

    // Input details for all students
    inputStudentDetails(students, n);

    // Display details of all students
    printf("\nDetails of all students:\n");
    displayStudentDetails(students, n);

    // Sort students based on marks in descending order
    sortStudentsByMarks(students, n);
    printf("\nStudents sorted by marks (descending):\n");
    displayStudentDetails(students, n);
```

```c
    // Find and display the student with the highest marks
    struct Student topStudent = findTopStudent(students, n);
    printf("\nStudent with the highest marks:\n");
    printf("Name: %s, Age: %d, Marks: %.2f\n", topStudent.name, topStudent.age, topStudent.marks);

    return 0;
}

// Function to input details of students
void inputStudentDetails(struct Student students[], int n) {
    for (int i = 0; i < n; i++) {
        printf("\nEnter details for student %d:\n", i + 1);
        printf("Name: ");
        scanf("%s", students[i].name);  // No spaces allowed in names, can be replaced with fgets if needed
        printf("Age: ");
        scanf("%d", &students[i].age);
        printf("Marks: ");
        scanf("%f", &students[i].marks);
    }
}

// Function to display details of students
void displayStudentDetails(struct Student students[], int n) {
    for (int i = 0; i < n; i++) {
        printf("Name: %s, Age: %d, Marks: %.2f\n", students[i].name, students[i].age, students[i].marks);
    }
}
```

```
// Function to sort students by marks in descending order
void sortStudentsByMarks(struct Student students[], int n) {
    struct Student temp;
    for (int i = 0; i < n - 1; i++) {
        for (int j = i + 1; j < n; j++) {
            if (students[i].marks < students[j].marks) {
                // Swap the students
                temp = students[i];
                students[i] = students[j];
                students[j] = temp;
            }
        }
    }
}

// Function to find the student with the highest marks
struct Student findTopStudent(struct Student students[], int n) {
    struct Student topStudent = students[0];
    for (int i = 1; i < n; i++) {
        if (students[i].marks > topStudent.marks) {
            topStudent = students[i];
        }
    }
    return topStudent;
}
```

```
Enter the number of students: 3

Enter details for student 1:
Name: stuti
Age: 12
Marks: 10

Enter details for student 2:
Name: kriti
Age: 16
Marks: 10

Enter details for student 3:
Name: shruti
Age: 13
Marks: 9

Details of all students:
Name: stuti, Age: 12, Marks: 10.00
Name: kriti, Age: 16, Marks: 10.00
Name: shruti, Age: 13, Marks: 9.00

Students sorted by marks (descending):
Name: stuti, Age: 12, Marks: 10.00
Name: kriti, Age: 16, Marks: 10.00
Name: shruti, Age: 13, Marks: 9.00

Student with the highest marks:
Name: stuti, Age: 12, Marks: 10.00
```